

Graduate Project: Retail Data Mining & Business Intelligence (CS633 Project)

Deliverables

1. Code file, and it should be related to the Output. Many instances I've seen in assignments where the code file does not match Output. I hope this is NOT the case for the Project.
2. Data Visualizations (all necessary graphs requested should be shown)
3. Communication clearly specifying what you are doing, why you are doing it, and I do want to read insights, and recommendations.

Project Due: July 30th (Thursday) by 6pm

We will not be accepting project delivery beyond this date.

Data Mining - CS633

Scenario

You are hired as a Data Mining Consultant for a retail company. Your task is to analyze customer transactions to uncover insights that improve revenue, marketing strategies, and operational decisions.

Objective

Use data mining, OLAP, and data analysis techniques to extract actionable insights from the dataset.

Dataset

File: *Retail_Transactions_12K.csv* provided in Blackboard with Project file

Part 1: Data Cleaning & Understanding

Load the dataset, inspect missing values, duplicates, and data types. Clean the dataset accordingly.

Reflection:

- What data quality issues did you find?
- How might poor data quality affect decisions?

Part 2: Data Transformation

Create new features such as Age Group and Revenue per transaction. Encode categorical variables and normalize numeric data.

Reflection:

- Why is transformation necessary?
- Which transformation had the biggest impact?

Part 3: Data Warehouse & OLAP

Identify facts and dimensions. Create a data cube using `pivot_table` or `groupby`. Perform roll-up, drill-down, slice, and dice operations.

Reflection:

- Which dimension provides the most insight?
- What patterns appear after aggregation?

Part 4: Data Generalization

Convert age into groups and dates into months/quarters.

Reflection:

- How does generalization simplify analysis?

Part 5: Data Visualization

Create histogram (Income), scatter plot (Income vs TotalAmount), and bar chart (Sales by Category).

Reflection:

- What trends do you observe?
- Which visualization is most informative?

Part 6: Correlation Analysis

Compute correlation matrix and identify strong relationships.

Reflection:

- Which variables are strongly related?
- Are any relationships surprising?

Part 7: Frequent Pattern Mining

Prepare transaction data and apply Apriori. Compute support, confidence, and lift.

Reflection:

- Which patterns are most frequent?
- Why are these useful?

Part 8: Association Rules

Generate association rules and evaluate them.

Reflection:

- Which rules are useful?
- Which rules are misleading?

Part 9: Classification (Decision Tree)

Build a decision tree model to predict Returned or High Spender.

Reflection:

- What features are most important?
- How reliable is the model?

Part 10: Business Insights & Recommendations

Provide actionable insights.

Questions:

- What products should be bundled?
- Which customers are most valuable?
- What drives returns?
- How can revenue be increased?